Gennoor Tech

TRAIN. INNOVATE. BUILD.

5-DAY DEVELOPER BOOTCAMP · 40 HOURS

AI Development Framework

Build production AI apps with LangChain, FastAPI, Streamlit, RAG & Agents

5	40	10+	6
DAYS	HOURS	FRAMEWORKS	AI APPS BUILT

Course Overview

This 5-day hands-on developer bootcamp covers the modern AI application development stack — from LLM APIs and orchestration frameworks through RAG pipelines, AI agents, and full-stack deployment. Participants build 6 production-ready AI applications using LangChain, Semantic Kernel, Hugging Face, FastAPI, Streamlit, Gradio, and enterprise cloud platforms. Every framework is compared head-to-head so developers can make informed architecture decisions.

The AI framework landscape is overwhelming. This bootcamp makes it clear. You'll build the same app in multiple frameworks, understand the tradeoffs, and leave with the ability to pick the right tool for any AI project. From simple chatbot APIs to multi-agent RAG systems to full-stack AI-powered web apps — all with production patterns, not toy examples.

Who Is This For?

Python Developers
Backend engineers adding AI capabilities to applications

Full-Stack Engineers
Developers building end-to-end AI-powered products

AI Engineers
Engineers evaluating and selecting AI frameworks

Platform Teams
Teams building internal AI platforms and tooling

Learning Outcomes

1. **Build with LLM APIs** — integrate Azure OpenAI, Claude API, Gemini API, and open-source models (Ollama) into applications with proper error handling and streaming
2. **Master AI orchestration frameworks** — build with LangChain, LangGraph, Semantic Kernel, and CrewAI — understand when to use each and framework migration patterns
3. **Implement production RAG** — chunking strategies, embedding models, vector databases (FAISS, Chroma, Azure AI Search), hybrid search, and reranking
4. **Build AI-powered web apps** — full-stack development with FastAPI backends, Streamlit and Gradio frontends, WebSocket streaming, and authentication
5. **Design AI agents and automation** — tool-calling agents, MCP servers, multi-agent systems, and workflow automation with N8N and Power Automate
6. **Deploy and scale AI apps** — Docker containerization, cloud deployment (Azure, AWS), API rate limiting, caching, monitoring, and cost optimization

Prerequisites

Python programming (intermediate), experience with REST APIs and JSON, basic Git knowledge. Familiarity with at least one web framework (Flask, Django, or FastAPI) is helpful.

5-Day Curriculum

Entirely hands-on — every session builds a working application. Schedule and session order are fully customizable based on your organization's needs and time zones.

01 LLM APIs & Prompt Engineering for Developers

LIVE CODING + HANDS-ON LABS

SESSION A

Working with LLM APIs

- Azure OpenAI API: chat completions, streaming, function calling
- Anthropic Claude API: messages, tools, long-context patterns
- Google Gemini API: multimodal inputs, grounding
- Lab: Build a multi-provider LLM gateway with fallback logic
- Open-source models via Ollama: Llama, Mistral, Qwen locally

SESSION B

Advanced Prompt Engineering in Code

- System prompts, structured outputs (JSON mode), guardrails
- Prompt templates, chaining, and dynamic few-shot selection
- Token management, cost tracking, and rate limit handling
- Lab: Build a document analysis API with structured JSON output
- Lab: Build a code review assistant with streaming responses

02 LangChain, Semantic Kernel & Orchestration

LIVE CODING + FRAMEWORK COMPARISON LABS

SESSION A

LangChain & LangGraph Deep Dive

- LangChain: chains, prompts, output parsers, memory
- LangChain tools: web search, database, API connectors
- LangGraph: stateful graph workflows, conditional routing
- Lab: Build a research assistant with LangChain + tools
- Lab: Build a multi-step approval workflow with LangGraph

SESSION B

Semantic Kernel, CrewAI & Comparison

- Semantic Kernel: plugins, planners, .NET and Python SDKs
- CrewAI: multi-agent crews, roles, tasks, delegation
- AutoGen: multi-agent conversation patterns
- Lab: Build the same app in LangChain vs Semantic Kernel
- Framework decision matrix: when to use which and why

03 RAG Pipelines & Vector Databases

LIVE CODING + HANDS-ON LABS + ARCHITECTURE DESIGN

SESSION A

RAG Architecture & Implementation

- RAG fundamentals: chunking, embedding, indexing, retrieval
- Chunking strategies: fixed, recursive, semantic, document-aware
- Embedding models: OpenAI, Cohere, open-source (sentence-transformers)
- Lab: Build a RAG pipeline for enterprise document Q&A
- Lab: Compare chunking strategies on the same dataset

SESSION B

Vector Databases & Hybrid Search

- FAISS, Chroma, Pinecone, Weaviate — comparison and selection
- Azure AI Search: hybrid search, semantic ranking, filters
- Reranking with Cohere and cross-encoders
- Lab: Build a hybrid search RAG system with Azure AI Search
- Lab: Build a multi-document chat with citation tracking

04 Full-Stack AI Apps — FastAPI, Streamlit & Gradio

LIVE CODING + HANDS-ON LABS + DEPLOYMENT

SESSION A

FastAPI for AI Backends

- FastAPI: async endpoints, WebSocket streaming, background tasks
- Authentication, rate limiting, and API key management
- MCP Server development: tools, resources, prompts protocol
- Lab: Build a production AI API with FastAPI + Docker
- Lab: Build an MCP server connecting AI to a SQL database

SESSION B

Streamlit, Gradio & Frontend Integration

- Streamlit: rapid AI app prototyping with chat interfaces
- Gradio: model demos, file upload, multimodal interfaces
- Connecting frontends to FastAPI backends
- Lab: Build a document Q&A app with Streamlit + RAG backend
- Lab: Build a multimodal AI demo with Gradio (image + text)

05 Agents, Automation & Capstone Deployment

CAPSTONE PROJECT + ARCHITECTURE REVIEW + PRESENTATIONS

SESSION A

AI Agents & Workflow Automation

- Tool-calling agents with function calling and MCP
- Multi-agent collaboration with CrewAI and LangGraph
- Workflow automation: N8N, Power Automate, Copilot Studio
- Lab: Build an automated research-to-report agent pipeline
- Lab: Connect AI agents to enterprise systems via MCP

SESSION B

Capstone: Full-Stack AI Application

- Team project: Design and build a complete AI application
- Backend (FastAPI) + Frontend (Streamlit) + RAG + Agent
- Docker containerization and cloud deployment
- Monitoring, logging, and cost optimization patterns
- Team presentations, architecture review, and next steps

Flexible & Customizable: Session order, timing, depth, and industry focus can be tailored to your organization's specific requirements. Content is regularly updated to reflect the latest AI developments.

Tools & Frameworks Covered

All frameworks used hands-on with live coding and comparative labs

LangChain | LangGraph | Semantic Kernel | CrewAI | AutoGen | Hugging Face | FastAPI | Streamlit | Gradio | MCP Servers | Azure OpenAI | Claude API | Gemini API | Ollama | FAISS / Chroma | Azure AI Search | N8N | Docker | Python

What You Take Home

🚀 6 AI Applications
Complete source code — LLM gateway, research assistant, RAG Q&A app, MCP server, Streamlit app, multi-agent pipeline

📐 Framework Decision Matrix
Detailed comparison: LangChain vs Semantic Kernel vs CrewAI vs AutoGen — features, performance, ecosystem, enterprise fit

🔧 AI App Starter Templates
Production-ready boilerplate for FastAPI + LangChain + Docker, Streamlit + RAG, and MCP server projects

📋 Architecture Patterns Guide
Reference for RAG patterns, agent architectures, API design, caching, streaming, and cost optimization strategies

Certification of Completion

Gennoor Tech — Certificate of Completion

Upon completing all 5 days (40 hours), each participant receives a **Gennoor Tech Certificate of Completion** in **"AI Development Framework"**. Digitally signed by the Founder & Director. Validates proficiency in AI orchestration frameworks, RAG implementation, full-stack AI app development, agent design, and production deployment.

Recommended Next Certifications

- **Microsoft AB-100** — Agentic AI Business Solutions Architect
- **Microsoft AI-103** — Azure AI App & Agent Developer Associate
- **AI-3026** — Develop AI Agents on Azure
- **GitHub GH-300** — GitHub Copilot Certification

Designed & Delivered By

Jalal Ahmed Khan

Founder & Director, Gennoor Tech Private Limited

AB-100 Certified Agentic AI Architect, Senior AI Consultant, and Microsoft Certified Trainer (MCT) with 14+ years of experience. Built and open-sourced MCP servers, RAG pipelines, and multi-agent systems. Delivered AI development programs for EY, Boeing, IBM, Microsoft, and 80+ organizations across 6+ countries.

16+ active Microsoft certifications including AI-103, AZ-400, AI-300, PL-400, GH-300, and GH-900. GitHub projects include Azure SQL MCP Server, prompt engineering labs, and banking document pipelines. **M.E. Mechanical Engineering**, Mumbai University.

Delivery Options

🏢 **On-Site** — customized at your office | 💻 **Virtual Live** — interactive via Teams/Zoom | 🌍 **International** — GCC, Africa, APAC, UK

Ready to Build
AI Applications?

Contact us to discuss a customized bootcamp
for your organization or register for the next open batch.

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